Assessing the Risk Factors of Under-Five Child Mortality: A Study Based on BDHS 2022

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Outline

- **Introduction**
- **Data and variables**
- **Methodology**
- **Analysis and Resutls**
- **Discussion and Conclusion**

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Background of study

- Child mortality, also known as mortality under five years of age, refers to the probability that a child born in a specific year will die before reaching the age of five, expressed per 1,000 live births (WHO, 2023).
- Being one of the crucial indicator to navigate the early health status of a child (UNICEF, 2023), child mortality is somewhat directly associated with one of the fundamental goals of SDG (UN, 2015).

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- Looking upon the world context, the global mortality rate for under five years of age has dropped by 59% since 1990 (WHO, 2023). Noticeable progress has been made in Bangladesh as well, dropping from 143.8 in 1990 to 31 in 2022 out of 1000 newborns (BDHS, 1993, 2022).
- However, it is necessary to keep reviewing the newly trends and fluidity of various socio-demographic factors to reach the optimum rate proposed by SDG which is 25 out of 1000 newborns (UN, 2015).

Literature Review

- Socio-demographic factors such as woman's education level, wealth index, preceding birth interval in regional disparities were found significant (Haq et al., 2020). Usage of contemporary contraceptives came out influencing mortality rates (Khan et al., 2023).
- Cultural traits such as early marriage had a role to play as well. Mother-child interaction based variables such as sex of the child, birth order, mother's age at birth were figured out to be significant in some studies (Hossain et al., 2022; Khan and Awan, 2017).

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- Several studies explored the regional disparities on micro level i.e district level and temporal region, some deciphered the gender disparities (Islam and Akter, 2023). Even the impact of serendipitous calamities such as COVID-19 were studied by many (Ahmmed et al., 2021).
- Pre-birth complication and histories to navigate cause specific death (Mazumder et al., 2024) and advanced machine learning techniques was used to decompose several socio-demographic factors (Rahman and Rahman, 2025).

Objective of the study

- To determine the factors that influence the under-five child mortality in Bangladesh.
- To determine the prevalence the under-five child mortality in Bangladesh.

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Source of data

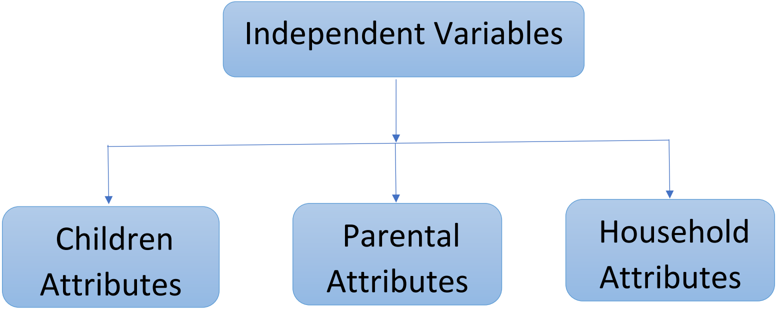
- Data for this study predominantly came from the Bangladesh Demographic and Health Survey (BDHS) 2022.

Dependent Variable

- There is 1 dependent variable in this study, which is "Child mortality status".
- Let Y be a binary response where,

$$Y = \begin{cases} 1 & \text{The child is death at or before age 5,} \\ 0 & \text{The child is not death at or before age 5.} \end{cases}$$

Independent variables



Children Attributes

- Sex of child
- Birth order
- BirthWeight
- Breastfeeding
- Preceding birth interval
- Mode of delivery
- Place of delivery
- Birth attendant during delivery

Parental Attributes

- Mother's Education
- Father's Education
- Mother's Occupation
- Mother age at child birth
- Antenatal Care (ANC)
- Media exposure

Household Attributes

- Division
- Residence
- Wealth index

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Methodology

- **Univariate analysis:**
- **Bivariate analysis:**
  - Pearson's Chi-square (χ^2) Test of Independence
- **Logistic Regression Model:**
  - Binary Logistic Regression Model
  - Multilevel Logistic Regression Model
  - Odds Ratio
  - Level of Significance and P-value
  - ROC Curve

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Results of bivariate table

Table: Bivariate Table of children Attributes

Characteristics	Child Survival		P-Value
	Dead N(%)	**Alive N(%)**	
Sex of child			
Male	149(3.28%)	4378(96.72%)	0.224
Female	118(2.72%)	4195(97.27%)	
Birth order			
First	96(2.86%)	3275(97.14%)	0.054
Second	77(2.57%)	2934(97.43%)	
3rd or 3rd+	92(3.76%)	2362(96.24%)	
Birth Weight			
Low birth weight	30(5.85%)	483(94.14%)	<0.001
Normal birth weight	31(1.15%)	2664(98.84%)	
Over birth weight	72(4.49%)	3379(95.51%)	

Characteristics	Child Survival		P-Value
	Dead N(%)	**Alive N(%)**	
Breastfeeding			
Yes	93(1.89%)	4827(98.11%)	<0.001
No	113(4.42%)	3746(95.58%)	
Preceding birth interval			
Short	29(4.89%)	575(95.11%)	0.019
Normal	139(2.86%)	4702(97.13%)	
Mode of delivery			
Cesarean section	27(1.89%)	2351(98.11%)	<0.001
Vaginal	118(4.42%)	2898(95.58%)	
Place of delivery			
Home	72(3.71%)	1880(96.29%)	<0.001
Public Hospital	33(3.60%)	886(96.40%)	
Private Hospital	38(1.54%)	2475(98.46%)	
Birth attendant during delivery			
Trained	67(2.02%)	3257(97.97%)	<0.001
Untrained	78(3.76%)	2002(96.24%)	

Table: Bivariate Table of parental Attributes

Characteristics	**Child Survival**		**P-Value**
	Dead N(%)	**Alive N(%)**	
Mother's Education			
No education	18(3.22%)	531(96.78%)	
Primary	71(3.47%)	1988(96.54%)	0.155
Secondary	149(3.12%)	4595(96.88%)	
Higher	29(1.97%)	1459(98.02%)	
Father's Education			
No education	57(4.11%)	1330(95.89%)	
Primary	91(3.42%)	2584(96.58%)	0.001
Secondary	90(2.97%)	2927(97.02%)	
Higher	23(1.40%)	1600(98.60%)	
Mother's Occupation			
Working	171(2.78%)	5988(97.22%)	0.101
Not working	95(3.55%)	2584(96.45%)	

Characteristics	Child Survival		P-Value
	Dead N(%)	Alive N(%)	
Mother age at child birth			
$<$20 year	55(3.19%)	1672(96.81%)	0.886
$20-30$ year	144(2.92%)	4803(97.08%)	
>30 year	36(3.02%)	1182(96.98%)	
Antenatal Care (ANC)			
<4	82(2.12%)	3803(97.88%)	0.749
$>=4$	24(1.95%)	1217(96.05%)	
Media exposure			
Yes	127(2.62%)	4717(97.38%)	0.048
No	139(3.49%)	3856(96.51%)	

Table: Bivariate Table of Household Attributes

Characteristics	Child Survival		P-Value
	Dead N(%)	Alive N(%)	
Division			
Barishal	18(3.21%)	552(96.80%)	0.213
Chattogram	59(3.08%)	1866(96.92%)	
Dhaka	56(2.52%)	2160(97.48%)	
Khulna	18(2.14%)	868(97.86%)	
Mymensingh	23(3.08%)	727(96.92%)	
Rajshahi	26(2.83%)	881(97.17%)	
Rangpur	38(3.92%)	925(96.08%)	
Sylhet	27(4.45%)	593(95.55%)	
Residence			
Urban	70(2.95%)	2316(97.05%)	0.861
Rural	196(3.03%)	6257(96.97%)	

Characteristics	Child Survival		P-Value
	Dead N(%)	Alive N(%)	
Wealth index			
Poorest	81(4.41%)	1768(95.59%)	<0.001
Poor	66(3.58%)	1772(96.42%)	
Middle	51(2.78%)	1769(97.22%)	
Rich	33(1.97%)	1668(98.03%)	
Richest	35(2.14%)	1596(97.86%)	

Results of multilevel logistic regression

Table: Results of multilevel logistic regression analysis of under-five child Mortality

Variables	Adj odds ratio	P-Value	[95% Conf. Interval]
Division			
Mymensingh	(Ref.)		
Barishal	0.444	0.275	(0.104,1.904)
Chittagong	0.116	0.004	(0.027,0.506)
Dhaka	0.168	0.014	(0.040,0.699)
Khulna	0.451	0.259	(0.113,1.798)
Rajshahi	0.359	0.190	(0.078,1.662)
Rangpur	0.968	0.962	(0.264,3.551)
Sylhet	0.233	0.049	(0.051,0.997)
Wealth Index			
Poorest	(Ref.)		
Poorer	0.879	0.450	(0.630,1.227)
Middle	0.713	0.062	(0.500,1.016)
Richer	0.511	0.193	(0.257,1.351)
Richest	0.567	0.236	(0.401,15.721)

Variables	Adj odds ratio	P-Value	[95% Conf. Interval]
Mother's Occupation			
Not working	(Ref.)		
working	0.744	0.498	(0.317,1.749)
Mother's education			
No education	(Ref.)		
primary	0.975	0.924	(0.591,1.612)
Secondary	0.899	0.663	(0.559,1.447)
Higher	0.957	0.031	(0.302,0.945)
Father's education			
No education	(Ref.)		
primary	4.552	0.027	(1.192,17.386)
Secondary	2.020	0.323	(0.500,8.157)
Higher	0.957	0.962	(0.158,5.811)
Media exposure			
No	(Ref.)		
Yes	0.804	0.962	(0.633,1.022)

Variables	Adj odds ratio	P-Value	[95% Conf. Interval]
Mode of delivery			
Vaginal	(Ref.)		
Caesarean section	0.307	0.009	(0.125,0.746)
Birth order			
First	(Ref.)		
Second	0.845	0.270	(0.628, 1.138)
3rd or 3rd+	1.264	0.106	(0.951,1.681)
Birth weight			
Low	(Ref.)		
Normal	0.113	<0.001	(0.051,0.250)
Over	0.428	0.153	(0.134,1.368)
Breastfeeding			
No	(Ref.)		
Yes	0.052	<0.001	(0.023,0.113)

Variables	Adj odds ratio	P-Value	[95% Conf. Interval]
Preceding birth interval			
Short	(Ref.)		
Normal	0.597	0.309	(0.221,1.614)
Sex of child			
female	(Ref.)		
male	2.117	0.045	(1.016,4.414)
Place of delivery			
Home	(Ref.)		
Public Hospital	1.080	0.905	(0.306,3.817)
Private Hospital	1.071	0.880	(0.284,4.047)
Birth attendant during delivery			
Trained	(Ref.)		
untrained	2.288	0.089	(0.882,5.933)
Intercept	0.358	0.473	(0.022,5.885)
Random Cluster Effect	0.4834		(0.304,0.766)

Results of Intracluster correlation coefficient

- The **ICC** of this model is 0.122 and 95% CI is (0.092 ,0.207).

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Discussion

- Focusing on examining the prevalence and associated factors of under-five mortality, this study utilizes multilevel binary logistic model that considers various factors such as division, mother's education, father's education, wealth index, birth order, and birth weight etc.
- Both maternal and paternal education attainment was found to be significantly associated with under-five child mortality rate. The more educative parents a child has, the less likely they become vulnerable to mortality exposure.

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- Exclusive breastfeeding and opt for cesarean section lessens the mortality rate, our finding suggest which is concurrent with other prior studies as well.
- Another consistent finding is normal birth weight reducing mortality risk.
- Gender and regional disparity based findings were interesting. According to our study, male childrens are more vulnerable to mortality than of female childs. A specific division, Mymensingh, came out as the most vulnerable whereas previous study points the arrow at Sylhet.

Conclusion

- Since parental education is a key to reduce child mortality, imperative policies should be more focused on advertising quality education for parents.
- More promotion advocating C-section, exclusive breastfeeding should be vocalized since they significantly lessen the mortality prevalence.
- To incorporate with the SDG' goal 3.2, attenuating the mortality rate to meet the optimum of 25 out of 1000 newborn should be the goal.
- Anticipating that this study's findings contribute to conceptualizing the national goal of achieving the minimal child mortality rate by 2030, aligning with the broader vision for 2041.

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Thank You